

Self-Learning Assignment Report

Electrical Wiring Control Systems for SumoBlue

MIME 497 — Senior Capstone

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Skill Learned and Motivation

For my self-learning assignment, I took on the challenge of learning wiring and electrical systems as applied directly to our Sumobot capstone project. Prior to this assignment, I had little to no formal background with electrical diagrams, calculating component power requirements, or using PCB boards. This skill develops several core attributes of engineering, including determining system compatibility and direct implementation.

I chose this topic because our team's Sumobot required a reliable and correctly rated power system to ensure our offensive and defensive design could come to life. I also took on the system control side of writing the Arduino code for this project, so I needed to work closely alongside the electronics to ensure all connections would be correct. Without a properly planned electrical system, we risked damaging motor drivers, servos, or the Arduino. These failures would set our project back with shipping delays and rework.

Evidence of Self-Learning

My research process involved the following activities (see References for full citations):

- Reviewed datasheets for every component to determine operating voltage and current draw (see Table 1).
- Watched tutorials on Arduino car assembly and the Flysky FS-I6X transmitter (needed to learn how to integrate the FS-I6X into the control code):
 - “RC Robot Car — RC Controls and Arduino” [2]: <https://www.youtube.com/watch?v=BACBNgaCnJU>
 - “Ultimate Arduino RC Car Build Using FS-i6X Transmitter!” [3]: <https://www.youtube.com/watch?v=I-AhMTsfvmE>
- Researched power distribution block wiring and how to apply crimps for Anderson Powerpole and Dupont connections:
 - Anderson Powerpole crimping [4]: <https://www.youtube.com/watch?v=QzLvdR6X81k&t=137s>
 - Dupont crimping [5]: <https://www.youtube.com/watch?v=mdD9NaCWuJ8>
 - Power distribution board overview [6] (drone-style PDB; used as a reference for understanding general operation before purchasing a dedicated 12 V PDB): <https://www.youtube.com/watch?v=9GFGGKaUNOk>

Table 1: Component Power Requirements

Component			Supply age	Volt-	Current Draw / Max- imum Allowance	Power Source
Left Wheel Driver	Motor		5 V		43 A maximum	5 V UBEC #1
Right Wheel Driver	Motor		5 V		43 A maximum	5 V UBEC #1
Motor Encoder			5 V		10 mA maximum	5 V UBEC #2
45 kg-cm Claw Servo			7.4 V		5.2 A	7.4 V UBEC #1
Bottom Ramp Servos ($\times 2$)			7.4 V		5.2 A each	7.4 V UBEC #2
Arduino Board			7–12 V (barrel jack)		30–50 mA	Direct from PDB
Motors ($\times 2$)			12 V		5.5 A	Direct from PDB
3S LiPo Battery			11.1 V / 12.6 V charged	nominal fully	5200 mAh capacity	Main supply

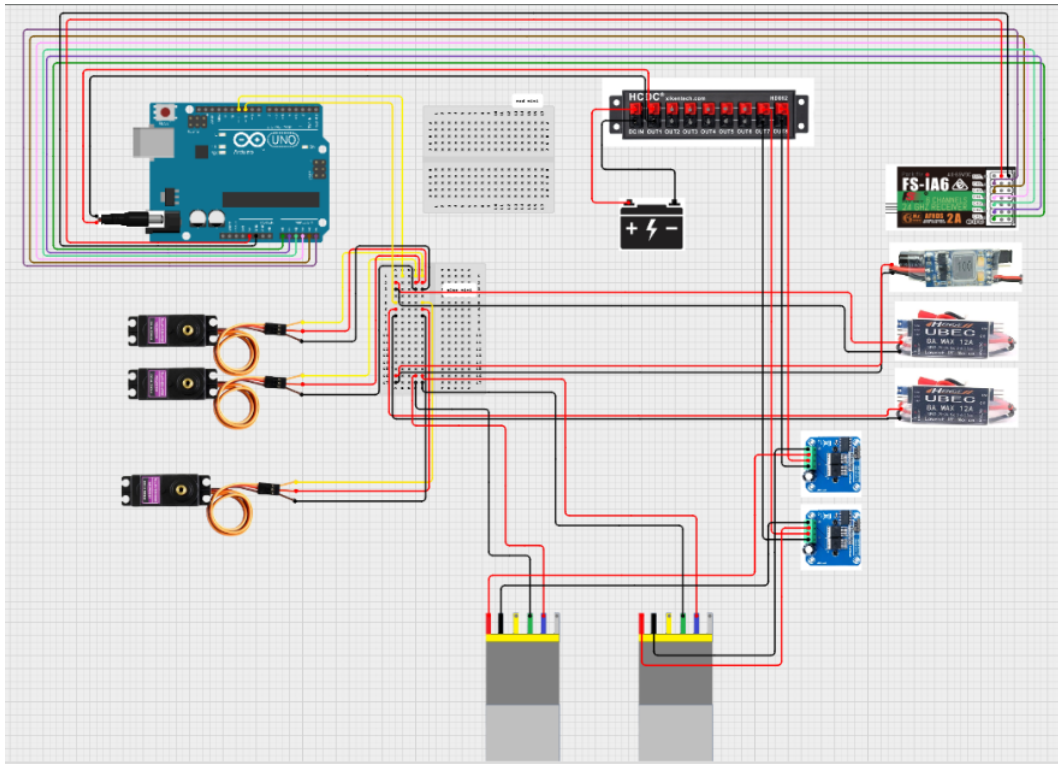


Figure 1: Full wiring diagram for the Sumobot electrical system. The complete physical system was assembled by the author; the diagram was reproduced in Tinkercad by teammate Bella Sifuentez. Power flows from the 3S LiPo through the power distribution block to each branch described in Section 2.

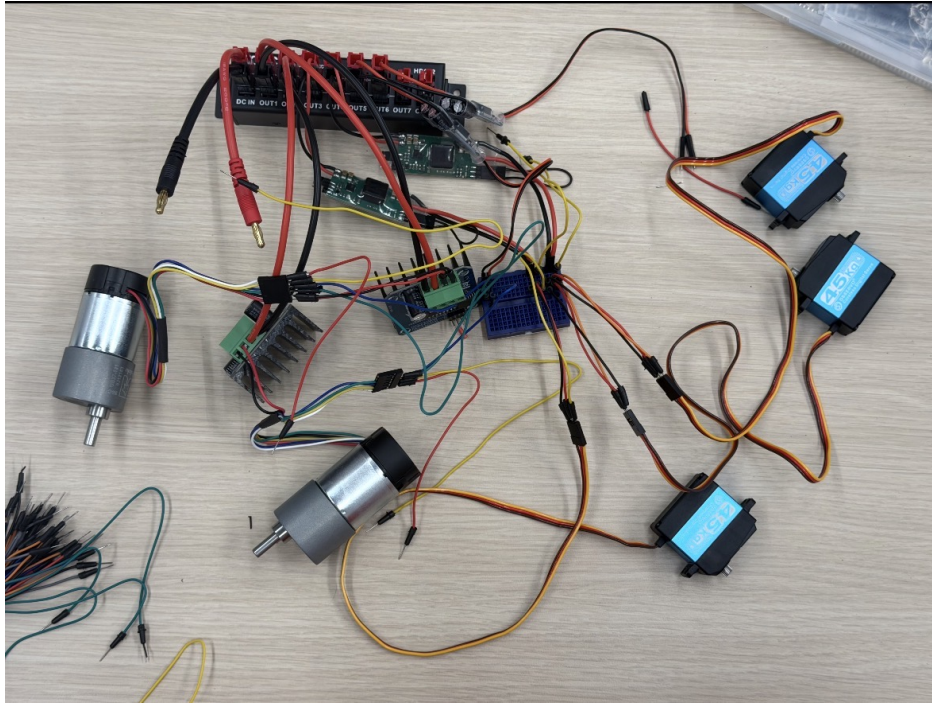


Figure 2: Initial test of the wiring system with all subsystems powered but prior to Arduino integration.

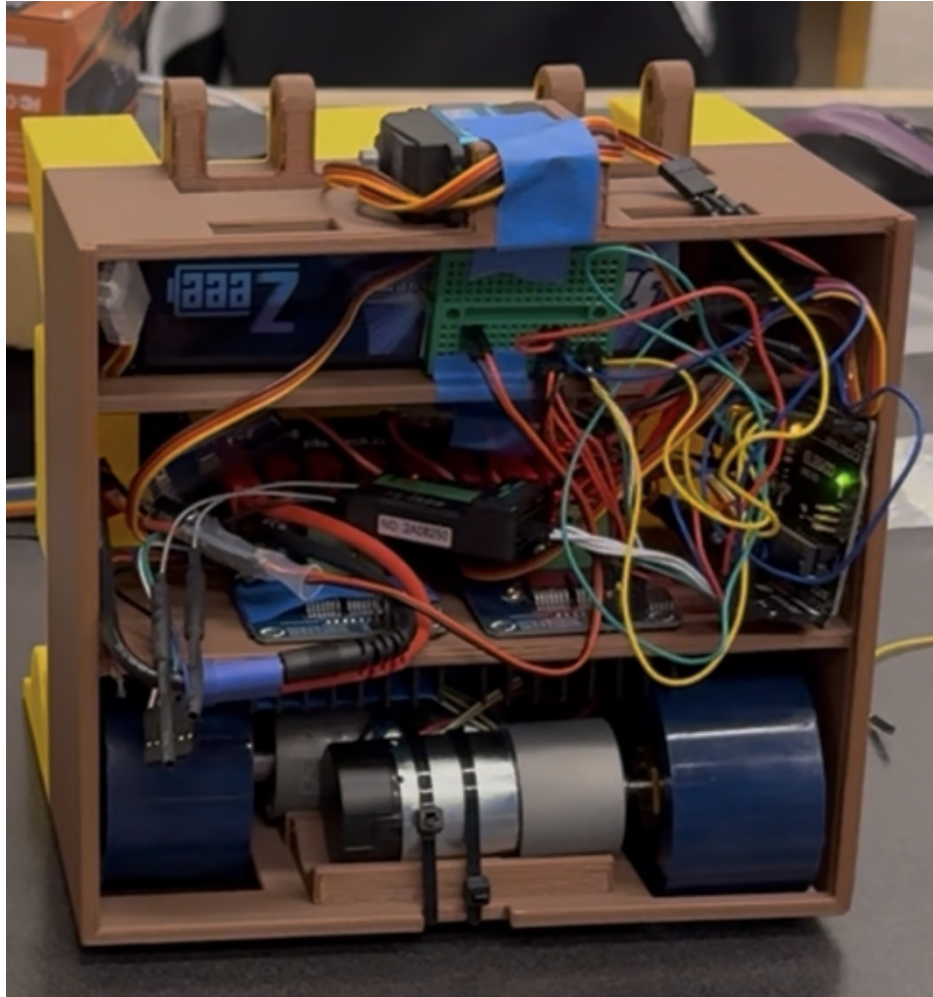


Figure 3: Functional test of the complete system installed inside the printed chassis.

Application of Skills and Results

After completing my research, I designed and built the complete wiring harness for the Sumobot. The power system developed flows as such:

1. **3S LiPo Battery (12 V nominal)** connects directly to a **power distribution block (PDB)**, which provides multiple parallel output rails from a single battery input. Learning to use the PDB was itself a new skill, as it required understanding how to safely crimp wires into terminals and calculate the total current and voltage budget for all connections.
2. **5 V — Motor Drivers:** Two 5 V UBECs step the battery voltage down to 5 V to power the logic inputs (LWMP and RWMP pins) of both motor drivers. This is key to determining motor polarity, allowing the robot to move both forwards and backwards.
3. **5 V — Motor Encoders:** A dedicated third 5 V UBEC powers the encoders on both drive motors, enabling variable speed adjustments for each motor independently.
4. **7.4 V — Claw Servo:** A 7.4 V UBEC powers the high-torque 45 kg-cm servo responsible for actuating the top claw of the Sumobot.

5. **7.4 V — Ramp Servos:** A second 7.4 V UBEC powers the two servos that articulate the bottom ramp of the Sumobot.
6. **Arduino Uno (Barrel Jack):** A barrel-jack connection from the PDB supplies voltage directly to the Arduino. Through research I learned that the Arduino has an onboard regulator that can accept up to 12 V and step it down to the required 5 V operating voltage.
7. **Motor Drivers — 12 V Direct:** Both motor drivers receive 12 V directly from the PDB to drive the DC motors at full voltage when commanded by signals from the Arduino.

In addition to the power connections, I coordinated all signal wiring between the Arduino and every actuator and sensor. This required cross referencing the Arduino Uno pinout with the code our team developed to ensure each PWM output and encoder interrupt pin was mapped to the correct physical connection.

Results

The first test of the assembled electrical system did not go exactly as planned. After getting everything set up, we realized there was a significant amount of wire management that needed to be addressed. Our solution was to implement a custom PCB and remove all non location dependent connections onto their own circuit board. This has not yet been implemented due to time constraints caused by shipping delays.

Beyond wire management, the system was also not fully operational on the first test. All servo subsystems were running as expected, but the drivetrain was not completely working. Specifically, the right wheel operated normally while the left wheel did not move at all. From there, we ran a full diagnostic with a multimeter to determine where the issue was arising and whether it was a hardware or software problem. After measuring the signal and supply voltages, we could confirm that every signal was being correctly sent to the left motor driver and that it was receiving a steady 12 V from the battery. This helped us deduce that the left motor driver had failed and was no longer operational.

Since that initial test, all subsystems have been connected and the code and power system are working well together. One minor issue remains: the servos zero out at startup, but this can be manually resolved by attaching the moment arms after the system has powered on.

Reflection

Overall, this was both a valuable and necessary learning experience for this project. Gaining hands on experience with wiring, voltage and amperage limits, and linking code to a physical wiring system was exactly what I had hoped to get out of this capstone. Coming into the project, I already had a strong background in CAD and initially expected to focus primarily on chassis design. However, being able to apply both skill sets effectively throughout this project has made it the most well rounded and rewarding engineering experience I have had so far.

References

- [1] Project component budget sheet. Each linked product includes a datasheet with amperage and voltage specifications.
<https://docs.google.com/spreadsheets/d/1RPeXTg2Bp0CdMD9FqeGtkzJmvUubi6fw3Zj4hoNW51Y/edit?usp=sharing>
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- [Online]. Available: <https://www.youtube.com/watch?v=BACBNgaCnJU>. Accessed: Apr. 19, 2026.
- [3] Robotics Lab, “Ultimate Arduino RC Car Build Using FS-i6X Transmitter!”, *YouTube*, Apr. 18, 2025. [Online]. Available: <https://www.youtube.com/watch?v=I-AhMTsfvmE>. Accessed: Apr. 19, 2026.
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- [6] Project Blue Falcon, “Power Distribution Board Overview,” *YouTube*, Jan. 13, 2017. [Online]. Available: <https://www.youtube.com/watch?v=9GFGGKaUNOk>. Accessed: Apr. 19, 2026.